RISHABH TRIPATHI

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Summary

Third-year B.Tech CSE (AI & Robotics) student at VIT Chennai (CGPA 9.01/10). Thirteen pull requests merged into projects I do not own, seven of them into Mantine (32k stars) and T3 Code (57k stars) since late October, each with a root cause and regression test. Builds distributed and local-first systems in Rust, Python and TypeScript: a desktop engine that OCRs and embeds images locally, an edge memory engine that escalates conflicts, and a serverless AWS study platform.

Education

Vellore Institute of Technology, Chennai **Aug 2024 – Aug 2028**
B.Tech, Computer Science & Engineering (AI & Robotics) *CGPA 9.01 / 10.00*

Competitive Programming

LeetCode: 100+ problems solved in C++. Weekly Contest rating 1,842 — top 4.01% globally, rank 15,234 of 380,000.

Open Source Contributions

Mantine | *mantinedev/mantine, 32k stars* **3 merged | 4 in review**
React component library | TypeScript *Oct 2026 – Present*

- Fixed `Combobox.HiddenInput` dropping falsy values: the prop is typed `Primitive | Primitive[] | null` but the implementation used a truthiness check, so a form submitted `count=` instead of `count=0`. #9359
- Fixed `<BarChart type="waterfall">` writing computed ranges onto the caller's data objects during render, permanently corrupting consumer state. #9365
- Also in review: a dropped `useForm.clearFieldError` update, where the field error was removed from the form state but the error node was never cleared from the DOM. #9385

T3 Code | *pingdotgg/t3code, 57k stars* **4 merged | 4 in review**
AI coding agent | TypeScript, Effect *Oct 2026 – Present*

- Fixed the Forgejo pull-request listing returning closed-unmerged rows in the merged tab and vice versa, because Forgejo's API has no merged state; both tabs burned their page limit on rows the client then discarded. #24772
- Made the Pi MCP extension write atomic by staging and renaming into place: the CLI loads that file from disk, and a non-atomic write left a concurrent reader with a missing file 54,612 times and an empty one 4,510 times across 43,000 reads. #32231
- Also in review: a Windows rename that fails with a sharing violation when a concurrent reader holds the extension file open. #34074

Experience

Ecera System Private Ltd. **Jun 2026 – Aug 2026**
Software Developer Intern | Remote

- Built a job portal end to end: schema and service layout, session-based authentication with role checks, and a Jenkins CI/CD pipeline that builds and deploys to a Linux VPS.
- Contributed to an LMS platform: refactored shared UI components for keyboard and screen-reader access, added CAPTCHA to public forms, and built real-time dashboard views.

SSoC (Social Summer of Code) **Jun 2026 – Aug 2026**
Project Admin, Traceon and AlgoForge | Remote

- Set architecture direction, ran code review and mentored contributors for two repositories I own (Traceon, 10 stars / 21 forks; AlgoForge, 9 stars / 13 forks).
- Shipped 17 merged pull requests into AlgoForge: rate limiting, schema validation on every input, atomic XP updates, leaderboard indexes, and a TTL cache that replaced an N+1 query pattern with three queries.
- Closed a data leak in the leaderboard API, where raw Mongo ObjectIds let anyone harvest users by ID; profiles now resolve through a random public id. Removed about 1,050 lines of duplicated UI and 20 unused packages, taking frontend lint from 51 warnings to zero.

GSSoC (GirlScript Summer of Code) **Apr 2026 – Aug 2026**
Open-Source Contributor | Remote

- Merged 5 pull requests across three GSSoC projects run by the piyushdotcomm organisation: Editron, VeriCred and LifeQuest.
- Diagnosed an Editron collaboration defect where the Yjs binding was torn down and rebuilt on every keystroke because live content was listed as an effect dependency, flooding the network with token requests, and opened a fix with a regression test. #460

Projects

tessera: Local-First Search Engine for Images and Documents (in development) github.com/rishabhx29/tessera
Rust, Tauri, ONNX Runtime, SQLite (FTS5 + sqlite-vec), notify | MIT

- Built a Rust workspace, a Tauri shell over a core engine, that indexes screenshots, PDFs, DOCX and text files on the machine and answers one query across all of them, with no account, no uploads and no telemetry.
- Indexer pipelines per format sit behind one trait, so a new format is a new implementation rather than a change to the store; search fuses three arms: FTS5 trigram, SigLIP for images, MiniLM for text, via RRF.

- Published benchmarks before any claims: the first configuration failed the latency budget, so OCR was A/B tested against three engines and Windows.Media.Ocr was kept at 35 ms p50 against RapidOCR's 2,820 ms, with the failing run kept on record.

edgemem: Disagreement-Durable Edge Memory Engine github.com/rishabhx29/edgemem

Python 3.12 | Qdrant Edge | WebSockets | pytest

- Designed an offline-first memory engine where a claim is an attributed assertion, never an overwritten fact: two claims that cannot both be true both survive with their attribution, and the engine escalates rather than choosing a winner.
- Every query returns one labelled verdict (ANSWERED_LOCALLY, CONFLICTED, UNRESOLVED_CLOUD_REQUIRED, CORRECTED) carrying the authority ladder, the reason a device may not fetch an answer, and its byte cost.
- Measured runtime performance instead of assuming it: a blocking index build stalls concurrent queries at 120k points but not at 50k, a hard process kill loses every unsynced write, and the vendor's snapshot path drops sparse vectors, which is why the engine has a write-ahead log.

lecturelens: Serverless AI Study Companion github.com/rishabhx29/lecturelens

AWS Lambda, API Gateway, SQS, DynamoDB, S3, Cognito, SAM | React, Vite

- Split the application into microservices behind API Gateway (upload, indexing, chat and a Study Pack worker) and served the frontend from a private S3 bucket over an API Gateway HTTPS host, with no public object access.
- Moved Study Pack generation off the request path onto an SQS queue with a Lambda worker and DynamoDB status polling, so a long generation never runs into the API Gateway 29-second timeout.
- Grounded answers in the document: PDFs are chunked and indexed, retrieval runs by cosine similarity with page citations, and the Bedrock Converse API is optional, leaving a deterministic local study engine as the zero-cost default.

mitra: Offline Voice Assistant for Windows github.com/rishabhx29/mitra

Electron, React 19, Three.js, TypeScript, Express | Vosk, Ollama

- Shipped a desktop assistant that runs with no API keys: on-device speech recognition through Vosk, a local LLM on Ollama with a roughly 50 ms decision tier for fast intent routing, and a permission gate the user controls for every tool call.
- Covered it with 767 unit tests across 42 files and 8 Playwright end-to-end tests, with every flow working in a mock mode so the demo needs no models and no credentials.
- Routed unknown requests through a fixed chain (Start-Menu index, DNS-verified site guess, LLM inference, web search), so no utterance dead-ends.

Sober: On-Device Screen-Time Blocker for Android github.com/rishabhx29/Sober

Kotlin, Jetpack Compose, Accessibility API, Room | MIT

- Built a screen-time blocker that declares no INTERNET permission and ships no account, analytics or third-party SDK, and that closes only the distracting part of an app, such as Instagram Reels or YouTube Shorts, rather than the whole app.
- Enforcement runs in a foreground service, so limits survive screen-off and reboot. Covered it with 565 unit tests, and the project was merged into `awesome-android-apps` by its maintainer.

Technical Skills

Languages: TypeScript, JavaScript (ES2022), Python 3.12, C++17, Java, Kotlin, SQL, Bash | Rust (ownership and borrowing, traits, generics, async, Tauri)

Frontend: React 19, Next.js 15/16, Tailwind CSS v4, React Flow, Three.js (WebGL, custom shaders), Framer Motion, GSAP, Zustand

Backend and APIs: Node.js, FastAPI, Django/DRF, Express, REST API design, WebSockets, Redis, PostgreSQL, MongoDB, DynamoDB, Prisma, Supabase, RLS

Cloud and DevOps: AWS (Lambda, API Gateway, S3, SQS, Cognito, DynamoDB, Bedrock), Docker, GitHub Actions, Jenkins, Vercel, Git

System Design: caching with TTL and in-flight de-duplication, write-ahead log, conflict detection and escalation, sharding, atomic file replacement, message queues and backpressure, idempotency, concurrency, rate limiting, OAuth 2.0, JWT, RBAC

AI and data: RAG (chunking, embeddings, cosine retrieval), hybrid search with RRF, ONNX Runtime, SigLIP, MiniLM, prompt-injection detection, local LLMs (Ollama, qwen2.5), on-device speech-to-text, vector search (Qdrant, HNSW)

Links: linkedin.com/in/rishabh-tripathi-728a77317 | github.com/rishabhx29 | rishabhx29.me | leetcode.com/u/Rishabh2903

Awards and Certifications

- Smart India Hackathon 2025: built and demoed a full-stack prototype inside a 36-hour window.
- GSSoC 2026 and SSoC 2026 certifications; Full-Stack Web Development (Udemy); Generative AI Workshop.
- GitHub Pull Shark and Pair Extraordinaire badges; 31 merged pull requests, 13 of them into upstream projects he does not maintain.